

Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Finds 2 nd derivative and sets up an inequality	AO3.1a	M1	$\frac{dy}{dx} = -2xe^{-x^2}$
	Obtains correct first derivative	AO1.1b	A1	$\frac{d^2y}{dx^2} = -2e^{-x^2} + 4x^2e^{-x^2}$
	Obtains second derivative correct from 'their' first derivative	AO1.1b	A1F	$-2e^{-x^2} + 4x^2e^{-x^2} < 0$ $4x^2 - 2 < 0$
	Deduces correct final inequality (could use set notation)	AO2.2a	A1	$-\frac{\sqrt{2}}{2} < x < \frac{\sqrt{2}}{2}$
(b)	Uses trapezium rule	AO1.1a	M1	$\int_{0.1}^{0.5} e^{-x^2} dx \approx \frac{0.1}{2}(e^{-0.01} + e^{-0.25})$ $+ 2(e^{-0.04} + e^{-0.09} + e^{-0.16})$ ≈ 0.3611
	Trapezium rule entries all correct	AO1.1b	A1	
	Finds correct value	AO1.1b	A1	
(c)	References area being completely within concave section So...	AO2.4	E1	$[0.1, 0.5] \subset \left[-\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right]$ $\therefore$ area is completely within concave section
	Trapezia all fall completely underneath the curve therefore underestimate (only award this mark if previous E1 has been awarded)	AO2.4	E1	Hence trapezia lie below curve and give an under-estimate for the area
(d)	Uses suitable rectangle to obtain over-estimate	AO3.1a	B1	Using a rectangle with the left hand edge the same height as the curve will produce an over-estimate
	Explains that this rectangle lies above the curve	AO2.4	E1	Area of rectangle =
	Constructs rigorous mathematical argument about accuracy, which leads to required result Only award if they have a completely correct solution, which is clear, easy to follow and contains	AO2.1	R1	$0.4 \times e^{-0.01^2} = 0.396...$ $\therefore 0.36 < A < 0.40$ So $A = 0.4$ to 1 dp

no slips.			
Total 12 marks			

Q2.

(a) $h = 1$

PI

B1

$$f(x) = \frac{1}{x^2 + 1}$$

$$I \approx \frac{h}{2} \{f(1) + f(5) + 2[f(2) + f(3) + f(4)]\}$$

$\frac{h}{2} \{f(1) + f(5) + 2[f(2) + f(3) + f(4)]\}$
OE summing of areas of the four 'trapezia'...

M1

$$\frac{h}{2} \text{ with } \{...\} = \frac{1}{2} + \frac{1}{26} + 2\left(\frac{1}{5} + \frac{1}{10} + \frac{1}{17}\right)$$

$$= 0.5 + 0.03(84...) + 2[0.2 + 0.1 + 0.05(88...)]$$

$$= 0.538(46...) + 2[0.358(82...)] = 1.256(108...)$$

OE Accept 2dp (rounded or truncated) for non-terminating decs. equiv.

A1

$$(I \approx) 0.628054... = \frac{694}{1105} = 0.628 \text{ (to 3sf)}$$

CAO Must be 0.628

SC for those who use 5 strips, max possible is B0M1A1A0

A1

4

(b) (i) $\int \left(x^{-\frac{3}{2}} + 6x^{\frac{1}{2}} \right) dx = \frac{x^{-\frac{1}{2}}}{-1/2} + \frac{6x^{\frac{3}{2}}}{3/2} (+ c)$

One term correct (even unsimplified)

M1

Both terms correct (even unsimplified)

A1

$$= -2x^{-0.5} + 4x^{1.5} (+ c)$$

Must be simplified

A1

3

$$(ii) \int_1^4 \left(x^{-\frac{3}{2}} + 6x^{\frac{1}{2}} \right) dx = [-2(4^{-0.5}) + 4(4^{1.5})] - [-2(1^{-0.5}) + 4(1^{1.5})]$$

Attempt to calculate $F(4) - F(1)$ where $F(x)$ follows integration and is not just the integrand

M1

$$= (-1 + 32) - (-2 + 4) = 29$$

Since 'Hence' NMS scores 0 / 2

A1

2

[9]

Q3.

(a) $h = 0.5$

$h = 0.5$ stated or used.

B1

$$f(x) = \sqrt{8x^3 + 1}$$

$$I \approx \frac{h}{2} \{f(0) + f(2) + 2[f(0.5) + f(1) + f(1.5)]\}$$

$$I \approx \frac{h}{2} \{f(0) + f(2) + 2[f(0.5) + f(1) + f(1.5)]\} \text{ OE}$$

M1

$$\frac{h}{2} \text{ with } \{\dots\} = \sqrt{1} + \sqrt{65} + 2(\sqrt{2} + \sqrt{9} + \sqrt{28})$$

$$= 1 + 8.06\dots + 2(1.41\dots + 3 + 5.29\dots)$$

$$= 9.0622\dots + 2 \times 9.7057\dots$$

OE Accept 1dp evidence. Can be implied by later correct work provided more than one term or a single term which rounds to 7.12

A1

$$(I \approx) 0.25[28.47\dots] (= 7.118\dots) = 7.12 \text{ (to 3sf)}$$

CAO Must be 7.12

A1

- (b) Stretch(I) in x -direction(II) scale factor 2 (III)
 Need (I) and either (II) or (III)

M1

Need (I) and (II) and (III)

More than 1 transformation scores 0/2

A1

- (c) $g(x) = \sqrt{(x-2)^3 + 1} - 0.7$
 $\sqrt{(x-2)^3 + 1} - 0.7$ or $\sqrt{(x-2)^3 + 1} + 0.7$
 or $\sqrt{(x+2)^3 + 1} - 0.7$ or $\sqrt{(x-2)^3 + 1} - 0.7$
 or their equivalents

M1

$$\sqrt{(x-2)^3 + 1} - 0.7 \quad \text{OE}$$

A1

$$g(4) = 2.3$$

2.3 OE

A1

3

Alternative:

(4, ...) on $y = g(x)$ comes from translating (2, 3) on $y = \sqrt{x^3 + 1}$
 from (2, ...) on $y = \sqrt{x^3 + 1}$

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from (2, 3) on $y = \sqrt{x^3 + 1}$

(A1)

(2, 3) after translation becomes (4, 2.3) so $g(4) = 2.3$

2.3 OE

(A1)

(3)

[9]

Q4.

- (a) $h = 1$

$h = 1$ stated or used. (PI by x -values 0, 1, 2, 3, 4 provided no contradiction)

B1

$$f(x) = \frac{2^x}{x+1}$$

$$\approx h/2\{\dots\}$$

$$\{.\} = f(0) + f(4) + 2[f(1) + f(2) + f(3)]$$

OE summing of areas of the 'trapezia'...

M1

$$\{.\} = 1 + \frac{16}{5} + 2\left(\frac{2}{2} + \frac{4}{3} + \frac{8}{4}\right)$$

$$= 1 + 3.2 + 2(1 + 1.33 \dots + 2)$$

OE Accept 1dp evidence. Can be implied by later correct work provided > 1 term or a single term which rounds to 6.43

A1

$$(I \approx) 0.5[4.2 + 2 \times 4.333 \dots] = 6.43 \text{ (to 3sf)}$$

CAO Must be 6.43

A1

- (b) Increase the number of ordinates

OE eg increase the number of strips.

E1

1

[5]

Q5.

- (a) $h = 0.25$

PI

B1

$$f(x) = \log_{10}(x^2 + 1)$$

$$I \approx h/2\{\dots\}$$

$$\{.\} = f(0) + f(1) + 2[f(0.25) + f(0.5) + f(0.75)]$$

OE summing of areas of the 'trapezia'

M1

$$\{.\} = \log 1 + \log 2 + 2\left[\log \frac{17}{16} + \log \frac{5}{4} + \log \frac{25}{16}\right]$$

OE Accept 1sf evidence

A1

$$= 0 + 0.3010... + 2(0.0263... + 0.0969... + 0.1938...)$$

$$= 0.3010... + 2(0.317058...) = 0.935147 \dots$$

$$(I \approx) 0.125 [0.935147 \dots] = 0.117 \text{ (to 3SF)}$$

CAO Must be 0.117

A1

4

(b) $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$

B1

1

(c) (i) $\log_{10} (10x^2) = \log_{10} 10 + \log_{10} x^2$

Condone missing bases for M mark.

Accept $\log x^2$ replaced by $2\log x$ in M1 line

M1

$$= 1 + 2 \log_{10} x$$

AG. Bases must be included or statement ' $\log_{10} 10 = 1$ ' given.

A1

Condone missing bases in (c)(ii) & (c)(iii)

2

(ii) $y = 1 + 2 \log_{10} x = \log_{10} (10x^2)$

PI

M1

Either $y = 2 \log_{10} (\sqrt{10}x)$ (to compare $y = 2 \log x$)

or both $y = \log_{10} x^2$ and $y = \log_{10} (\sqrt{10}x)^2$

Writing in correct form so that stretch details can be stated directly

A1

(Stretch) parallel to x -axis, sf $\frac{1}{\sqrt{10}}$ OE

B2 for correct direction and scale factor ACF

(B1 for correct exact scale factor ACF)

(or B1 for ' x -direction, scale factor $1/10$ ')

(or B1 for ' x -direction, scale factor $\sqrt{10}$ ')

Apply ISW if dec follows exact values.

(OE scale factor must be in exact form)

B2,1,0

4

(iii) $\log_{10} (10x^2) = \log_{10} (x^2 + 1)$

PI by $10x^2 = x^2 + 1$ or correct x

M1

$$(10x^2 = x^2 + 1, 9x^2 = 1 \text{ and since } x > 0) x = \frac{1}{3}$$

$$x = \frac{1}{3} \text{ OE stated or used; accept } \sqrt{\frac{1}{9}} \cdot \frac{1}{\sqrt{9}}$$

A1

(y-coordinate of P) $y = \log_{10} \frac{10}{9}$

Or $y = \log \left(\frac{1}{9} + 1 \right)$

PI by $3 \log \frac{10}{9}$ OE for the gradient of OP

A1

Gradient of OP = $3 \log_{10} \frac{10}{9} = \log_{10} \frac{1000}{729}$

$\log \frac{1000}{729}$; Accept 'a = 1000, b = 729'

A1

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[15]

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Q6.

(a) $h = 0.5$

PI

B1

$$f(x) = \sqrt{27x^3 + 4}$$

$$I \approx h/2 \{ \dots \}$$

$$\{ \dots \} = f(0) + f(1.5) + 2[f(0.5) + f(1)]$$

OE summing of areas of the 'trapezia'.

M1

$$\{ \dots \} = \sqrt{4} + \sqrt{95.125} + 2(\sqrt{7.375} + \sqrt{31})$$

$$= 2 + 9.7532... + 2(2.7156... + 5.5677...)$$

OE Accept 2dp rounded or truncated as evidence for surds

A1

$$(I \approx) 0.25 \times 28.32012... = 7.08 \text{ (to 3sf)}$$

Must be 7.08

A1

4

(b)

$$g(x) = \sqrt{27\left(\frac{1}{3}x\right)^3 + 4} = \sqrt{x^3 + 4}$$

*Any form which simplifies to $\sqrt{kx^3 + 4}$, $k \neq 27$, $k \neq 0$
or which simplifies to $x^3 + 4$*

M1

ACF

A1

2

[6]

Q7.

(a) $h = 0.5$

*$h = 0.5$ stated or used. (PI by x -values
0, 0.5, 1, 1.5, 2 provided no contradiction)*

B1

$$f(x) = \sin x$$

$$I \approx h/2\{\dots\}$$

$$\{.\} = f(0) + f(2) + 2[f(0.5) + f(1) + f(1.5)]$$

OE summing of areas of 'trapezia'..

M1

$$\{.\} =$$

$$0 + 0.90929.. + 2[0.4794.. + 0.84147.. + 0.99749..]$$

$$\{.\} = 0.90929.. + 2[2.318..] = \{ 0.90929.. + 4.636..\}$$

*Min. of 2dp values rounded or truncated.
Can be implied by later correct work
provided > 1 term or a single term which
rounds to 1.39*

A1

$$(I \approx) 0.25[5.546..] = 1.3865.. = 1.39 \text{ (to 3sf)}$$

CAO Must be 1.39

A1

4

- (b) Stretch(I) in y-direction(II) scale factor 2(III)

Need (I) and either (II) or (III)

M1

*All correct. Need (I) and (II) and (III)
[> 1 transformation scores 0/2]*

A1

2

(c) $\frac{\sin x}{\cos x} = \frac{1}{2}$; $\tan x = 0.5$

$$\frac{\sin x}{\cos x} = \tan x \quad \text{used to get } \tan x = k$$

*or identity $\cos^2 x + \sin^2 x = 1$ used to get either
 $\sin^2 x = p$ or to get $\cos^2 x = q$, (p and q must be
between 0 and 1)*

M1

$$\tan x = 0.5$$

Either $\tan x = \frac{1}{2}$ or $\cos x = \pm \sqrt{\frac{4}{5}} (= \pm 0.894..)$

or $\sin x = \pm \sqrt{\frac{1}{5}} (= \pm 0.447..)$

A1

$$x = \alpha \text{ or } \pi + \alpha \text{ where } \alpha = \tan^{-1}(k)$$

*Correct method to find 2nd angle. Any in wrong
ft quadrants then m0. In case of squaring
method candidates must also have rejected
the extra 'quadrants' for the m1. Condone
degrees or mixture*

m1

$$x = 0.464, 3.61$$

*Both. Condone > 3sf [0.463(6..), 3.60(5..or 6..)]
Accept pair of truncated values [0.463, 3.60]
Ignore any answers outside interval 0 to 6.28*

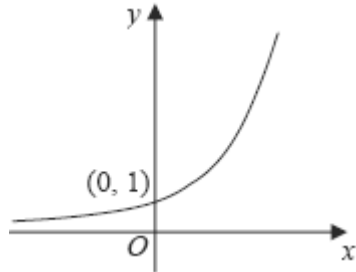
A1

4

[10]

Q8.

(a)



Shape with some indication of asymptotic behaviour in 2nd quadrant below pt of intersection with y-axis

B1

Only intersection is with y-axis at (0, 1) stated/indicated ... (accept 1 on y-axis as equivalent)

B1

2

(b) (i) $h = 0.5$
PI

B1

$$f(x) = 2^x$$

$$I \approx h/2\{\dots\}$$

$$\{\dots\} = f(0) + f(2) + 2[f(0.5) + f(1) + f(1.5)]$$

OE summing of areas of the 4 'trapezia'

M1

$$\begin{aligned} \{\dots\} &= 1 + 4 + 2(\sqrt{2} + 2 + \sqrt{8}) \\ &= 5 + 2 \times 6.2426\dots = 17.485\dots \end{aligned}$$

OE Accept 2 dp (rounded or truncated) as evidence for surds

A1

$$(I \approx) 4.3713\dots = 4.37 \text{ (to 3 sf)}$$

CAO Must be 4.37

SC for those who use 5 strips, max possible is B0M1A1A0

A1
4

- (ii) Increase the number of ordinates
OE

E1
1

- (c) Translation;
*Accept 'translat...' as equivalent
[T or Tr is NOT sufficient]*

B1;

$$\begin{bmatrix} -7 \\ 3 \end{bmatrix}$$

*B1 for each component of the vector.
Condone if the equiv 2 vectors are given.
Accept **full** equivalent to vector(s) in
words provided linked to 'translation/
move/shift' and **correct** directions.
(No marks if **different** transformations)*

B1;B1
3

- (d) $8 = 2^k + 3 \Rightarrow 2^k = 5$

Correct subst. and an attempted

rearrangement to $2^k = N$. PI by $k = \frac{\log 5}{\log 2}$

M1

$$k = \log_2 5$$

Accept $m = 2, n = 5$

A1
2

[12]

Q9.

- (a) $(y =) 1$

B1
1

- (b) $h = 0.2$

PI

B1

$$f(x) = 2^{4x}$$

$$I \approx h/2 \{...\}$$

$$\{.\} = f(0) + f(1) + 2[f(0.2) + f(0.4) + f(0.6) + f(0.8)]$$

OE summing of areas of the 'trapezia'..

M1

$$\{.\} = 1 + 16 + 2(2^{0.8} + 2^{1.6} + 2^{2.4} + 2^{3.2})$$

$$= 1 + 16 + 2(1.741.. + 3.031.. + 5.278.. + 9.1895..)$$

$$= [17 + 2 \times 19.24...]$$

OE Accept 2dp rounded or truncated evidence

A1

$$I = 5.55 \text{ (to 2 dp)}$$

Must be 5.55

A1

4

- (c) Stretch(I) in y -direction(II) scale factor $\frac{1}{8}$ (III)
- Need (I) **and** either (II) or (III)*

M1

Need (I) and (II) and (III)

A1

ALTn: Translation with an indication
that the translation is in the x -direction (B1)

$$\begin{bmatrix} 3 \\ 4 \\ 0 \end{bmatrix} \quad (\text{B1})$$

*Combination of **different**
transformations scores 0/2*

2

(d) $g(x) = 2^{4(x-1)} - \frac{1}{2}$

B1 for either $2^{4(x+1)} - \frac{1}{2}$ or for

$$2^{4(x-1)} + \frac{1}{2} \text{ or for } 2^{4x-1} - \frac{1}{2}$$

B2,1,0

At $Q, y = 0 \Rightarrow 2^{4(x-1)} = 2^{-1}$

Reaches a stage from which linear eqn can be stated directly eg an alternative stage is

$$4(x - 1) \log 2 = -\log 2$$

M1

$$\Rightarrow 4x - 4 = -1 \Rightarrow x = 0.75$$

NMS mark as 4 or 0

A1

4

(e) (i) $\log_a k = \log_a 2^3 + \log_a 5 - \log_a 4$
 $\log_a k = \log_a (2^3 \times 5) - \log_a 4$

One law of logs used

M1

A second law of logs used; could be

$$\log_a k = \log_a 2^3 + \log_a \left(\frac{5}{4}\right)$$

M1

$$\log_a k = \log_a \left(\frac{2^3 \times 5}{4}\right) = \log_a 10 \Rightarrow k = 1$$

CSO AG

A1

3

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(ii) $2^{4x-3} = \frac{5}{4}$ so

Equate y's, take logs (to any base) of both sides and apply 3rd law of logs.

$$(4x - 3) \log_{10} 2 = \log_{10} \frac{5}{4}$$

M1

$$x = \frac{3 \log_{10} 2 + \log_{10} \left(\frac{5}{4}\right)}{4 \log_{10} 2}$$

m1

$$x = \frac{\log_{10} 10}{4 \log_{10} 2} \text{ so } x = \frac{1}{4 \log_{10} 2}$$

A1

Altn $4x \log 2 = \log \left(\frac{5}{4} \times 2^3 \right)$

Rearrange correctly to $x = \dots$

Altn $4x \log 2 = \log 10$

In both cases, log term(s) must have same base and expressions must be in an exact form, ie not approx. dec. vals

CSO AG Must be clear evidence that base 10 is used, also be convinced

3

[17]

Q10.

(a) $h = 1.5$

PI

B1

$$f(x) = x^2 \sqrt{x^2 - 1}$$

Integral = $h/2 \{ \dots \}$

$$\{ \dots \} = f(1.5) + 2[f(3) + f(4.5)] + f(6)$$

For the M1 covered range must be 1.5 to 6
OE summing of areas of the three traps.

M1

$$\{ \dots \} = 2.51(5..) + 2[25.4(5..) + 88.8(4..)] + 212(.9..)$$

Check at least 3 sf values, rounded or truncated, or award if a combined value WRT 444 is seen or final answer is 333 or rounds to 333
Condone one numerical slip

A1

$$\text{Integral} = 0.75 \times 444.1 = 333 \text{ to 3sf}$$

Must have 333

Treat using 4 strips as a MR and mark with max of B0M1A1A1cao as follows:

$$h = 1.125 \text{ B0}$$

$$\{ \dots \}$$

$$= f(1.5) + 2[f(2.625) + f(3.75) + f(4.875)] + f(6) \text{ M1}$$

$= 2.51(5) + 2[16.7(2) + 50.8(2) + 113(.3)] + 212(.9)$ A1
 or award if a combined value WRT 577 is
 seen or final answer is 325 or rounds to
 325. Condone one numerical slip.
 Answer = 325 A1cao Must have 32

A1cao

4

- (b) Increase the number of ordinates

OE eg increase the number of strips

E1

1

[5]

Q11.

- (a) $h = 2$

PI

B1

$$g(x) = \sqrt{x^3 + 1}$$

$$I \approx h/2 \{ \dots \}$$

$$\{ \dots \} = g(0) + g(6) + 2[g(2) + g(4)]$$

OE summing of areas of the 'trapezia'.

Can award even if MR expression for $g(x)$
 but must be using from 0 to 6

M1

$$\{ \dots \} = 1 + \sqrt{217} + 2(3 + \sqrt{65})$$

$$1 + 14.73\dots + 2(3 + 8.06\dots)$$

OE Accept 2 dp evidence for surds

A1

$$(I \approx) 37.8554\dots = 37.86 \text{ (to 4 sf)}$$

Must be 37.86

A1

4

(b) $f(x) = \sqrt{(2x)^3 + 1} = \sqrt{8x^3 + 1}$

$$\sqrt{kx^3 + 1}, \quad k \neq 1 \text{ or } 0 \text{ or } f(x) = g(2x)$$

M1

Either form acceptable

A1

2

[6]

Q12.

$$h = 1$$

PI

B1

$$I = \frac{h}{2} \{ \dots \}$$

OE summing of areas of the three 'trapezia'

M1

$$\{ \dots \} = f(0) + f(3) + 2[f(1) + f(2)]$$

$$\{ \dots \} = \sqrt{3} + \sqrt{12} + 2[\sqrt{4} + \sqrt{7}]$$

$$(\Sigma trap = 1.866. + 2.3228. + 3.0549)$$

A1

$$(I \approx) \frac{1}{2} [5.19615. + 2 \times 4.64575 \dots]$$

$$= \frac{1}{2} [14.4876 \dots] = 7.2438 \dots = 7.244$$

CAO Must be 3 dp.

A1

[4]

Q13.

(a) (i) $h = 0.5$

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$$\text{Integral} = h/2 \{ \dots \}$$

PI

B1

$$\{ \dots \} = f(0) + 2\left[f\left(\frac{1}{2}\right) + f(1) + f\left(\frac{3}{2}\right)\right] + f(2)$$

OE summing of areas of the four traps.

M1

$$\{ \dots \} = 1 + 2[\sqrt{6} + 6 + 6\sqrt{6}] + 36$$

$$= 1 + 2 [2.449 \dots + 6 + 14.6969 \dots] + 36$$

Condone 1 numerical slip. Accept 3sf values if not exact.

A1

$$= 37 + 2 \times 23.146.. = 83.292...$$

$$\text{Integral} = 0.25 \times 83.292.. = 20.8 \text{ (3sf)}$$

CAO; must be 20.8

A1

4

- (ii) Relevant trapezia drawn on a copy of given graph
Accept single trapezium with its sloping side above the curve

M1

{Approximation is an} overestimate
Dep. on 4 trapezia with each of their upper vertices lying on the curve

A1

2

- (b) (i) Stretch (I) in x -direction (II)
Need (I) and one of (II), (III)
M0 if more than one transformation

M1

(scale factor) $\frac{1}{3}$ (III)

A1

2

(ii) $6^{3x} = 84$
PI

M1

$$\log_{10} 6^{3x} = \log_{10} 84$$

Take logs of both sides of $a^x = b$, PI by 'correct' value(s) later or $3x = \log_6 84$

M1

$$3x \log_{10} 6 = \log_{10} 84$$

Use of $3 \log 6^{3x} = 3x \log 6$ OE
or $3x = \log_6 84$ seen

m1

$$x = \frac{\lg 84}{3 \lg 6}$$

$$x = 0.82429\dots = 0.824 \text{ (to 3dp)}$$

*Must see that logs have been used before
any of the last 3 marks are awarded in
(b)(ii). Condone > 3dp*

A1

4

(c) $f(x) = 6^{x-1} - 2$

B1 For either $6^{x-1} + 2$ or for $6^{x+1} - 2$

B2,1

2

[14]

Q14.

$$h = 1$$

PI

B1

$$f(x) = \sqrt{2^x}$$

$$\text{Area} \approx h/2 \{ \dots \}$$

$$\{ \dots \} = f(0) + f(3) + 2[f(1) + f(2)]$$

OE summing of areas of the 'trapezia'..

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M1

$$\{ \dots \} = 1 + \sqrt{8} + 2(\sqrt{2} + 2)$$

OE

A1

$$(\text{Area} \approx) 5.3284\dots = 5.328 \text{ (to 3dp)}$$

CAO Must be 5.328

A1

4

[4]

Q15.

(a) $y_A = 3(2^0 + 1)$

Substituting $x = 0$ PI

M1

= 6

A1

2

(b) $h = 2$

PI

B1

Integral = $h/2 \{ \dots \}$

$\{ \dots \} = f(0) + 2[f(2) + f(4)] + f(6)$

OE summing of areas of the three traps.

M1

$\{ \} = 6 + 2[3 \times 5 + 3 \times 17] + 3 \times 65 = 6 + 2[15 + 51] + 195$

Condone 1 numerical slip {ft on (a) for f(0) if not recovered}

[Sum of 3 traps. = 21 + 66 + 246]

A1

Integral = 333

CAO

A1

4

(c) (i) $21 = 3(2^x + 1) \Rightarrow 2^x = 6$

AG (be convinced)

B1

1

(ii) $\log_{10} 2^x = \log_{10} 6$

Take ln or log₁₀ of both sides of $a^x = b$ or other relevant base if clear. The equation $a^x = b$ used must be correct.

M1

$x \log_{10} 2 = \log_{10} 6$

Use of $\log 2^x = x \log 2$ OE

m1

$x = \frac{\lg 6}{\lg 2} = 2.5849 \dots = 2.58 \text{ to 3sf}$

Both method marks must have been awarded.

A1

Q16.

(a) $h = 1$

PI

B1

$$\text{Integral} = \frac{h}{2} \{...\}$$

$$\{...\} = f(0) + f(4) + 2[f(1) + f(2) + f(3)]$$

OE summing of areas of the four trapezia.

$$[0.75+0.35+0.15+0.079...]$$

M1

$$= \left[1 + \frac{1}{17} + 2\left(\frac{1}{2} + \frac{1}{5} + \frac{1}{10}\right) \right]$$

Exact or to 3dp values Condone one numerical slip

A1

(b) Integral = 1.329

CSO. Must be 1.329

A1

4

Increase the number of ordinates

OE

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E1

1

[5]

Q17.

(a) (i) y -coordinate of A is $27 - 3^0 = 26$

M1A1

2

(ii) When $x = 3$, $y = 27 - 3^3 = 0 \Rightarrow B(3,0)$

AG; be convinced

B1

1

(b) $h = 1$

PI

B1

$\text{Area} \approx h/2\{\dots\}$
 $\{\dots\} = f(0) + f(3) + 2[f(1) + f(2)]$
OE summing of areas of the 'trapezia'..

M1

$\{\dots\} = "26" + 0 + 2(24 + 18)$
on (a)(i) ($\Sigma \text{trap} = "25" + 21 + 9$)

A1 ✓

(Area $\approx$) 55
on $[42 + 0.5 \times "(a)(i)"]$

A1 ✓

4

(c) (i) $\log_{10} 3^x = \log_{10} 13$
Takes $\ln$ or $\log_{10}$ on both or $x = \log_3 13$

M1

$x \log_{10} 3 = \log_{10} 13$
 $\frac{\lg 13}{\lg 3}$
Use of $\log 3^x = x \log 3$ or $\log_3 13 = \frac{\lg 13}{\lg 3}$ OE
(PI by $\log_3 13 = 2.335$ or better)

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m1

$x = \frac{\lg 13}{\lg 3} = 2.334717$
 $\dots = 2.3347$ to 4dp

Must show that logarithms have been used

A1

3

(ii) $\{k\} = 14$
Condone $y = 14$; Accept final answer 14 with only zeros after decimal point e.g. 14.000

B1

1

(d) (i) Translation;
'Translation' / 'translate(d)'
B0 if more than one transformation

B1

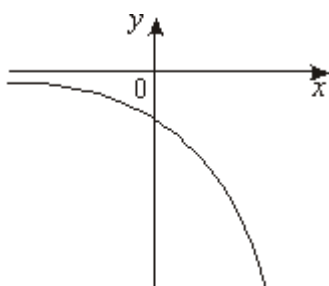
$$\begin{bmatrix} 0 \\ -27 \end{bmatrix}$$

Accept full equivalent in words provided
linked to 'translation/move/shift' and negative y-direction
(Note: B0 B1 is possible)

B1

2

(ii)



Correct shape (translation of given curve vertically downwards)

B1

Only point of intersection with coordinate axes is on negative y-axis and curve is asymptotic to the negative x-axis

B1

2

[15]

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Q18.

(a) {y-coordinate of A is} 2

B1

1

(b) (i) $h = 0.25$

B1

$$\text{Integral} = \frac{1}{2} \{ \dots \}$$

$$\{ \dots \} =$$

$$f(0) + 2\left[f\left(\frac{1}{4}\right) + f\left(\frac{1}{2}\right) + f\left(\frac{3}{4}\right)\right] + f(1)$$

$$\{...\} =$$

M1

$$2 + 4 + 2[(2.316.. + 2.732.. + 3.279(5..)]$$

A1 ✓

$$\{= 6 + 2 \times 8.3276..\} \{= 22.65(5...)\}$$

Condone one numerical slip
Accept values to 3sf (rnd or trunc)
ft answer from (a) if not "2"

$$\text{Integral} = 0.125 \times 22.655.. = 2.8319..$$

$$\text{Integral} = 2.83 \text{ to 3 sf}$$

CAO Must be 2.83
(NMS scores 0/4)

A1

4

- (ii) Relevant trapezia drawn on a copy of given graph

Accept relevant single trapezium with its
sloping side above the curve

M1

{Approximation is an}overestimate

A1

2

(c) $5 = 3^x + 1 \Rightarrow 3^x = 4$

B1

$$\log_{10} 3^x = \log_{10} 4$$

Takes $\ln$ or $\log_{10}$ on both sides of $3^x = k$, where $k > 0$

M1

$$x \log_{10} 3 = \log_{10} 4$$

Use of $\log 3^x = x \log 3$

m1

$$x = \frac{\lg 4}{\lg 3} = 1.26185 \dots$$

$$= 1.2619 \text{ to 4dp}$$

Accept 4dp or better
[If using T&I a full justification is
required; else M0m0A0]

A1
4

(d) $f(x) = 3^{-x} + 1$

B1
1

[12]

Q19.

(a) (i) $\sqrt{x} = x^{\frac{1}{2}}$
Accept $p = 0.5$

B1
1

(ii) $\int \sqrt{x} \, dx = \frac{x^{1.5}}{1.5} \{+c\}$
Index raised by 1

Correct ft on p . Condone missing '+c'

M1
A1ft
2

(iii) Area = $\int_1^4 \sqrt{x} \, dx$
Limits 1 and 4 PI

$$= \frac{4^{1.5}}{1.5} - \frac{1}{1.5}$$

$$F(4) - F(1)$$

B1
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M1

$$= \frac{14}{3}$$

Accept 4.66 or better

A1
3

(b) (i) $y = x^{\frac{1}{2}} \Rightarrow \frac{dy}{dx} = \frac{1}{2} x^{-\frac{1}{2}}$

Index (p – 1) ft

M1

When $x = 4$, $y'(4) = 0.25$

Attempt to find $y'(4)$

M1

When $x = 4$, $y = 2$

B1

Equation of tangent: $y - 2 = \frac{1}{4}(x - 4)$

accept other forms

A1

4

(ii) When $x = 0$, $y = 1$ $B(0, 1)$

Subs $x = 0$ and then $y = 0$ into equation of tangent. Pl

M1

When $y = 0$, $x = -4$ $A(-4, 0)$

*Correct ft y_B and x_A
(may be awarded as part of area calculation)*

A1ft

Area = $0.5(1)(4) = 2$

ft wrong sloping tangent and max of 1 further slip. Final answer must be +ve

A1ft

3

(c) Translation

'Translation'/'translate(d)'

B1

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

*Accept equivalent in words provided linked to 'translation/move/shift'
(Note: B0B1 is possible)*

B1

(d) $h = 1$

PI

B1

$$\text{Integral} = h/2 \{ \dots \}$$

$$\{ \dots \} = f(1) + 2[f(2) + f(3)] + f(4)$$

OE summing of areas of the three traps

M1

$$\{ \dots \} = 0 + 2(1 + \sqrt{2}) + \sqrt{3}$$

Condone 1 numerical slip

A1

$$\text{Integral} = \frac{1}{2} \{ 2(1 + 1.414\dots) + 1.732 \}$$

$$\text{Integral} = 0.5 \times 6.560\dots = 3.28 \text{ to 3sf}$$

Must be 3.28

A1CAO